

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 04

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Recap of Previous Lecture



Topic

Memory Addressing

Topic

Memory Access

Topics to be Covered



Topic

Microoperations



Topic : Address of Variable Size Instructions

ex:- byte addressable mem.

Program $\rightarrow$ start from add. 200 in mem.

Instns	size (in bytes)
I1	2 bytes
I2	3
I3	4
I4	3
I5	2

mem.	
200	I1
202	I2
205	I3
209	I4
212	I5
214	

Ans = 8

#Q. Consider the following program segment. Here R1 and R2 are the general purpose register. Assume that the content of memory location 2000 is 20. All numbers are in decimal. After the execution of this program the value of memory location 2000 is?

Instructions	Operations
MOV R1, #12	$R1 \leftarrow \#12$
MOV R2, (2000)	$R2 \leftarrow M[2000]$
SUB R2, R1	$R2 \leftarrow R2 - R1$
MOV (2000), R2	$M[2000] \leftarrow R2$
HALT	Stop

 $R1 = 12$ $R2 = 20$
8

Memory



Ans = 45

#Q. Consider the following program segment. Here R1 and R2 are the general purpose register. Assume that the content of memory location 3000 is ~~18~~23. All numbers are in decimal. After the execution of this program the value of memory location 3000 is?

Instructions	Operations
MOV R1, # 9 11	$R1 \leftarrow \#911$
MOV R2, (3000)	$R2 \leftarrow M[3000]$
ADD R2, R1	$R2 \leftarrow R2 + R1$
ADD R1, R2	$R1 \leftarrow R1 + R2$
MOV (3000), R1	$M[3000] \leftarrow R1$
HALT	Stop

R1 = ~~11~~ 45R2 = ~~23~~
34

Mem.

3000

~~23~~ 45

Ans = 21

#Q. Consider the following program segment. Here R1 and R2 are the general-purpose register. Assume that the content of memory location 1000 is 5. All the numbers are in decimal.

Instructions	Operations
MOV R1, (1000)	$R1 \leftarrow M[1000]$
MOV R2, #6	$R2 \leftarrow \#6$
LOOP: ADD R2, R1	$R2 \leftarrow R2 + R1$
DEC R1	$R1 \leftarrow R1 - 1$
BNZ LOOP	Branch on not <u>zero</u>
HALT	Stop

$R1 = \cancel{4} \cancel{3} 2$
10

$R2 = 6$
11
15 18
20
21

mem.

1000

5

The value of R2 at the end of program execution is?

#Q. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

Instruction	Operation	Instruction Size (no. of words)
MOV R1, (3000)	$R1 \leftarrow M[3000]$	2
MOV R2, (R3)	$R2 \leftarrow M[R3]$	1
ADD R2, R1	$R2 \leftarrow R2 + R1$	1
MOV (R3), R2	$M[R3] \leftarrow R2$	1
INC R3	$R3 \leftarrow R3 + 1$	1
DEC R1	$R1 \leftarrow R1 - 1$	1
BNZ LOOP	Branch on not zero	2
HALT	Stop	1

loop:

[illegible]

$$R3 = \begin{array}{ccc} \cancel{2000} & \cancel{2001} & \cancel{2003} \\ & \cancel{2002} & \cancel{2004} \\ & & 2005 \end{array}$$

$$R1 = \frac{10}{887} \approx 0.0112$$

if loop runs x times

$$\text{no. of mem. references for data} = 1 + 2 * x$$

loop runs 10 times $\Rightarrow x = 10$

1000	Inst ^{ns}
2000	100 110
2001	100 109
2002	100 108
2003	100 107
2004	100 106
2005	100 105
2006	100 104
2007	100 103
2008	100 102
2009	100 101
2010	100
	⋮
3000	10

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. The number of memory reference for accessing the data in executing the program completely is

A

10

B

11

C

20

D

✓ 21

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is:

A ✓ 100

B 101

C 102

D 110

#Q. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
1000	MOV R1, (3000)	$R1 \leftarrow M[3000]$	2
1008	MOV R2, (R3)	$R2 \leftarrow M[R3]$	1
1012	ADD R2, R1	$R2 \leftarrow R1 + R2$	1
1016	MOV (R3), R2	$M[R3] \leftarrow R2$	1
1020	INC R3	$R3 \leftarrow R3 + 1$	1
1024	DEC R1	$R1 \leftarrow R1 - 1$	1
1028	BNZ LOOP	Branch on not zero	2
1036	HALT	Stop	1

address for
word addressable

$$2 = 2 * 4B = 8B$$

$$1 = 4B$$

$$1 = 4B$$

$$1 = 4B$$

$$1 = 4B$$

$$1 = 4B$$

$$2 = 8B$$

$$1 = 4B$$

1000

1002

1003

1004

1005

1006

1007

1009

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is byte addressable and the word size is ^{4 Bytes}32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on the stack?

A 1005

B 1020

C ✓ 1024

D 1040

when interrupt occurs during execution of an instⁿ,
then address of next instⁿ is stored on stack as return add.

Ques) In prev. Questⁿ if mem. is word addressable, then return add. pushed onto stack — ?

Ans = 1006

#Q. Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and MEMORY [X] denotes the content at the memory location X.

Instruction	Semantics	Instruction Size (bytes)
MOV R1, (5000)	$R1 \leftarrow \text{MEMORY}[5000]$	4
MOV R2, (R3)	$R2 \leftarrow \text{MEMORY}[R3]$	4
ADD R2, R1	$R2 \leftarrow R1 + R2$	2
MOV (R3), R2	$\text{MEMORY}[R3] \leftarrow R2$	4
INC R3	$R3 \leftarrow R3 + 1$	2
DEC R1	$R1 \leftarrow R1 - 1$	2
BNZ 1004	Branch if not zero to the given absolute address	2
HALT	Stop	1

Assume that the content of the memory location 5000 is 10, and the content of the register R3 is 3000. The content of each of the memory locations from 3000 to 3010 is 50. The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is_____.



Topic : Calculating Address from Memory Size

Assume memory (byte addressable memory)

1B
1B
1B
1B

4 cells

$$\begin{aligned}\text{memory capacity} &= 4 * 1B \\ &= 4 \text{ Bytes}\end{aligned}$$

$$\text{memory capacity} = \text{no. of cells} * \text{size of content at each cell}$$

$$\text{no. of cells} = \frac{\text{mem. Capacity}}{\text{per cell content}}$$

ques mem. capacity = 64 bytes
byte addressable.

$$\begin{aligned} \text{no. of cells in memory} &= \frac{64}{1} \text{ ?} \\ \text{no. of bits for mem. add.} &= \underline{6 \text{ bits}} \text{ ?} \end{aligned}$$

Solⁿ

$$64 \text{ B} = \text{no. of cells} * 1 \text{ B}$$

$$\text{no. of cells} = \frac{64 \text{ B}}{1 \text{ B}}$$

$$= 64$$

$$= 2^6$$

ques.

mem capacity = 4K bytes
byte addressable mem.

$$\text{address size} = \underline{12} \text{ bits}$$

Solⁿ

$$\text{no. of cells} = \frac{4 \text{ KB}}{1 \text{ B}}$$

$$= 4 \text{ K} = 2^2 \cdot 2^{10}$$

$$= 2^{12}$$



$$\text{add.} = 12 \text{ bits}$$

#Q. Consider a memory with size 128 bytes. The address size to access memory, if memory is byte addressable, is 7 bits?

$$\begin{aligned}\text{no. of cells} &= \frac{128\text{B}}{1\text{B}} \\ &= 128 \\ &= 2^7 \Rightarrow \text{add.} = 7 \text{ bits}\end{aligned}$$

[NAT]



$$\text{no. of cells} = 16M = 2^4 \cdot 2^{20} = 2^{24} \Rightarrow \text{add} = 24 \text{ bits}$$

#Q. Consider a memory with size 16Mbytes. The address size to access memory, if memory is byte addressable, is 24 bits?

#Q. Consider a memory with size 2048 bytes. The address size to access memory, if memory is word addressable (1 word = 4 bytes), is 9 bits?

$$\text{no. of cells} = \frac{2048 \cancel{\text{B}}}{4 \cancel{\text{B}}} = \frac{2^{11}}{2^2} = 2^{11-2} = 2^9$$

↓
add. = 9 bits

#Q. Consider a memory with size 64Kbytes. The address size to access memory, if memory is word addressable(1 word = 2 bytes), is 15 bits?

$$\text{no. of cells} = \frac{64\text{KB}}{2\text{B}} = \frac{2^6 \cdot 2^{10}}{2^1} = 2^{15} \Rightarrow \text{add.} = 15 \text{ bits}$$

Ques) mem. add. size = 14 bits

byte addressable mem.

mem. capacity size = 16KB ?

Solⁿ

$$\text{no. of cells} = 2^{14}$$

$$\begin{aligned}\text{mem. capacity} &= 2^{14} * 1 \text{ B} \\ &= 2^{14} \text{ B} \\ &= 16 \text{ KB}\end{aligned}$$

Ques) mem add. length = 29 bits

word addressable mem.

word size = 64 bits = 8 bytes

mem capacity = 4 GB ?

Solⁿ

no. of cells = 2^{29}

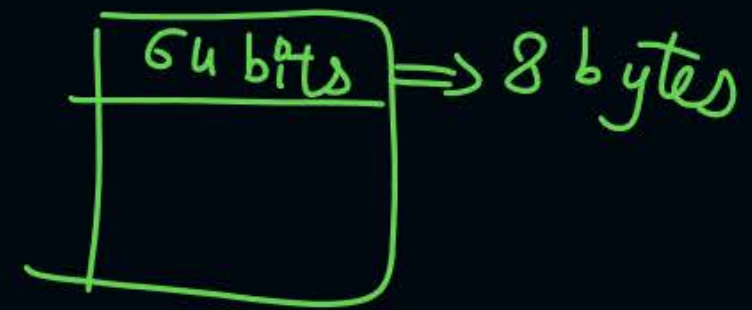
mem. capacity = $2^{29} * 8$ bytes

= $2^{29} * 2^3$ bytes

= 2^{32} bytes

= $2^{30} \cdot 2^2$ bytes

= 4 GB



b $\Rightarrow$ bits

B $\Rightarrow$ bytes



2 mins Summary



Topic

Microoperations





Happy Learning

THANK - YOU

